FREEDOM INTERNATIONAL SCHOOL

WORKSHEET - MCQ - SOLUTIONS

PHYSICS

CLASS XII

MOVING CHARGES AND MAGNETISM

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1. A voltmeter of resistance 2000 Ω, 0.5 V/div is to be converted into a voltmeter to make it to read 2V/div. The value of high resistance to be connected in series with it is

   (a) 6000 Ω (b) 4000 Ω (c) 5000 Ω (d) 1000 Ω

   Ans: (a) 6000 Ω

   I_g= V/(R + G)

   In first case: I_g = 0.5n/2000

   In second case: I_g = 2n/(2000+R)

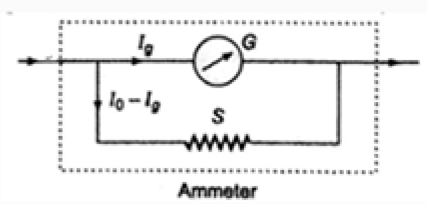
   Solving: R=6000 Ω

2. A galvanometer of resistance 25 Ω is shunted by a 2.5 Ω wire. The part of the total current that flows through the galvanometer is given by

   (a) $\frac{I}{I_0}=\frac{1}{11}$ (b) $\frac{I}{I_0}=\frac{3}{11}$ (c) $\frac{I}{I_0}=\frac{2}{11}$ (d) $\frac{I}{I_0}=\frac{4}{11}$

   Ans: (a) $\frac{I}{I_0}=\frac{1}{11}$

In the circuit G and S are in parallel, the potential difference across them is same, hence



$$I_g \times G = (I_0 - I_g) \times S$$

$$\Rightarrow \frac{I_g}{I_0} = \frac{S}{S+G}$$

Given, $S = 2.5\,\Omega,\ G = 25\,\Omega$

$$\therefore \frac{I_g}{I_0} = \frac{2.5}{25+2.5} = \frac{2.5}{27.5} = \frac{1}{11}$$

Take $I_g = I$

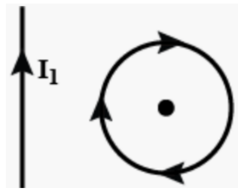
$$\therefore \frac{I}{I_0} = \frac{1}{11}$$

3. The ratio of voltage sensitivity V_s and current sensitivity I_s of a moving coil galvanometer is

(a) $\frac{1}{G}$ (b) $\frac{1}{G^2}$ (c) G (d) G^2

Ans: (a) $\frac{1}{G}$

4. In the given figure, the loop is fixed but straight wire can move.



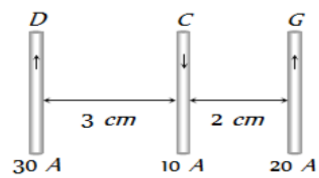
The straight wire will

(a) remain stationary (b) move towards the loop

(c) move away from the loop (d) rotate about the axis

Ans: (b) move towards the loop

5. Three long, straight parallel wires, carrying current, are arranged as shown in the figure. The force experienced by a 25 cm length of wire C is

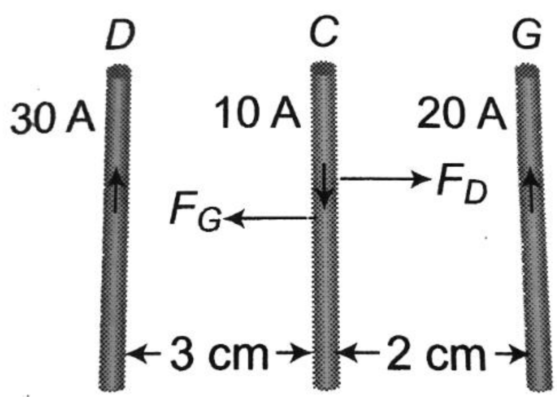


(a) 10^{-3} N (b) $2.5\text{x}10^{-3}$ N

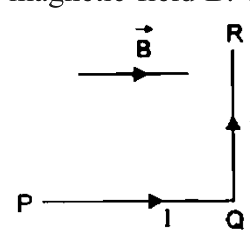
(c) zero (d) $1.5\text{x}10^{3}$ N

Ans: (c) zero

Force will be equal and opposite



6. A wire PQR is bent as shown in the figure and is placed in a region of uniform magnetic field B. The length PQ = QR = l. A current I ampere flows through the wire as shown. The magnitude of the force on PQ and QR will be



(a) Bil, 0 (b) 2 Bil, 0

(c) 0, Bil (d) 0, 0

Ans: (c) 0, Bil

7. The ratio of time period of an α particle to that of a proton circulating with same speed in the same uniform magnetic field is

(a) $\sqrt{2}:1$ (b) $1:\sqrt{2}$ (c) 1: 2 (d) 2: 1

Ans: (d) 2:1

The time period of revolution of a charged particle in a magnetic field is

$$T=\frac{2\pi m}{Bq}$$

For proton, $m_p=m, q_p=q$;

$$\therefore T_p=\frac{2\pi m}{Bq}$$

Now, for α -particle, $m_\alpha=4m$, $q_\alpha=2q$

$$\therefore T_\alpha=\frac{2\pi(4m)}{B(2q)}=2\left(\frac{2\pi m}{Bq}\right)$$

T_α/T_p= 2:1

8. If a particle is moving in a uniform magnetic field, then

(a) both momentum and total energy will change

(b) its momentum changes but total energy remains the same

(c) both momentum and total energy remain the same

(d) its total energy changes but momentum remains the same

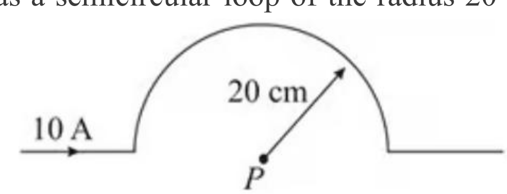
Ans: (b) its momentum changes but total energy remains the same

9. A current of 10 A is passing through a long wire which has a semicircular loop of the radius 20 cm as shown in the figure.

Magnetic field produced at the centre of the loop is

(a) 10π μT (b) 5π μT

(c) 4π μT (d) 2π μT



Ans: (b) 5π μT

$$B=\frac{1}{2}\left(\frac{\mu_0 I}{2r}\right)$$

10. The electrons in the beam of a television tube move horizontally from south to north. The vertical component of the earth's magnetic field points down. The electron is deflected towards

(a) west (b) no deflection (c) east (d) north to south

Ans: (c) east

For questions 11 to 15, two statements are given- one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

11. **Assertion:** If an electron is not deflected when moving through a certain region of space, then the only possibility is that no magnetic field is present in that region.
Reason: Force on an electron is directly proportional to the strength of the magnetic field.

Assertion is false and reason is true

12. **Assertion:** The magnetic field produced by a current carrying solenoid is independent of its length and area of cross-section.
Reason: Magnetic field within a very long solenoid is uniform.
Ans: B
13. **Assertion:** Galvanometer to ammeter conversion takes place by connecting a low value resistance in parallel with it.
Reason: The low value resistance increases the effective resistance and protects the galvanometer.
Ans: C
14. **Assertion:** A charge, whether stationary or in motion produces a magnetic field around it.
Reason: Moving charges produce only electric field in the surrounding space.
Ans: D
15. **Assertion:** The resistance of an ideal voltmeter should be infinite.
Reason: The lower resistance of a voltmeter gives a reading lower than the actual potential difference across the terminals.
Ans: A